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Trademarks pending: Plasma Agent · Plasma Agency
plasmaagentplatform.com · 2026-10

For those who build plasma process tools, and those who buy them — one physics engine, before anything is built

Every new source generation costs two or three prototypes; uniformity is tuned by experience; and the buyer does not know the process window until the tool is installed. **Plasma Agent Platform** is **software and a design service**: a physics engine that computes the ion and radical flux at the wafer from coil, electrode or tube geometry and operating conditions in **0.2 s**, with a twin on top that fits to measurement and recommends outcomes.

→ Makers

Geometry to outcome. Screen tens of thousands of geometries, compare two or three designs, get splitter specs and a process-window map.

tool makers · universities · RF component makers

← Buyers

Process target to source spec. Write the purchase specification on physics, and compare vendor proposals on the same yardstick.

device makers · vendor comparison

NAMES

Plasma Agent	The physics engine — Database · Source Modules · 0-D Global Model · Radial Profile. **Engine license**
Plasma Agency	The twin — Calibration · Process Models · Empirical Model & Recommender. **Twin + design service**
Agents	Three uses of the same engine — R&D · Process Optimization · Machine Control

ROADMAP · ONE ENGINE, THREE AGENTS

NOW

R&D Agent

Choose the design before building the source; buyers write the purchase spec. Patent filed.

NEXT

Process Optimization

Predict and recommend process results from a recipe. Calibrated on pilot process data.

LAST

Machine Control

Keep and correct the recipe at the tool. Joint development with tool makers.

Every step runs on the same physics engine and the same data model — cross-section data → 0-D characterization and source design → 2-D calibration → measurement calibration → twin → iterate (the core of the patent filing). The twin never controls the tool: it reads, recommends, and waits for approval.

SEVEN MODULES — THE ENGINE AND THE TWIN

Database Plasma Agent	Rate and EEDF tables generated at run time from public cross-section files (LXCat). Sources stay attached to results.	**Source Modules** Plasma Agent	Three ICP types · CCP · substrate bias · remote + diffuser — four source types, one interface, identical conditions.
0-D Global Model Plasma Agent	Mixed-gas, per-ion particle and power balances → Te, density, radicals. Metastables, gas heating, pulsed mode; 0.2 s per run.	**Radial Profile** Plasma Agent	Absorbed-power profile → radial density and temperature → wafer flux profile and non-uniformity. Basis for screening and purchase specs.
Calibration Plasma Agency	2-D results and measurements (impedance, V-I, probe, wafer) fitted only to the parameters each can inform. Physics and process layers kept apart.	**Process Models** Plasma Agency	Rate equations from plasma state to process result: plasma treatment · etch · deposition · strip. **Treatment (nitridation) validated on in-house data**; the rest in development.
Recommender Plasma Agency	Process-window map, sensitivity, next condition, pulsed recipes. Applied only after approval.		

WHO IT IS FOR · WHAT THERE IS, HOW YOU USE IT

RF system component makers

Load inductance, coupled resistance, efficiency, plasma potential and sheath impedance from geometry, gas, pressure and power in 0.2 s. Matcher range, supply design and pulse transients before delivery.

Plasma Agent engine license · calibrate device factors on your own V-I data

Universities and labs

Run-time generation from public cross-section files; antenna to radial profile in 0.2 s. Sources are recorded with each result, ready for teaching and papers.

Plasma Agent academic license · joint research through Plasma Agency

Tool makers — source development

Tens of thousands of geometries screened to two or three designs, splitter specs, a process-window map and a simulator package. Two to four prototypes per generation become one.

Plasma Agency design service → twin → joint tool integration

Device makers — process teams

Work backwards from the process target to the source spec. Purchase specification and vendor comparison before buying; recipe-to-result prediction after.

Plasma Agency purchase spec · vendor comparison → calibrated on wafer data

WHAT WE OFFER

Plasma Agent engine license	Four modules. Validation ratios and applicability grades stated.	**release frozen**
Plasma Agency twin	Engine + Calibration + Process Models + Recommender. Prediction and calibration history kept.	**v0.5 · treatment model validated**
Design service	Spec → scan → screening → designs + splitter spec + process-window map + package. For device makers: purchase spec and vendor evaluation.	**available now**
Reference source	A 300 mm N_2 treatment ICP source bundled with the twin — proof of "a source the twin designed".	**3 designs · after funding**

VALIDATED RANGE · VS LITERATURE AND 2-D

ICP design trends (density · uniformity · efficiency)	**±20 %**
Antenna electrics (L · coupled R · current)	**±10–25 %**
Mixed-gas Te / ion composition	**±0.3 eV / ×0.9–1.2**
CCP parallel-plate V · P · n_e (13.56–60 MHz)	**×1.0–1.3**
CCP low-pressure ion flux · energy · sheath width	**×0.7–1.5**
Substrate-bias mean ion energy	**×0.7–1.0**
Remote downstream density	**×0.5–0.9**
Treatment (nitridation) rate equation — 27 in-house points, same tool	**residual ≈12 %**

LIMITS, STATED PLAINLY

The 0-D model is strong on **trends and relative comparison**; absolute density carries a ×0.5–1.2 band. A purchase spec is therefore written as **ranges and conditions**, not a single number.

No source designed by the twin has yet been built and verified — **not yet.** Without a facility of our own, the first case comes from a pilot tool or a public fab.

Process Models: only treatment (nitridation) is validated; etch, deposition and strip are in development. CCP validation is for Ar; dual-frequency and high-pressure voltage-driven CCP are out of range.

Every number can be traced to an equation, a data source and a calibration record. Not overstating is the basis of trust.

Looking for one pilot to apply the design service on its next source, and RF component makers and labs who will try the engine first.

Send us your spec — we first tell you whether it is in range and computable. Details are shared under NDA.

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